

Semantic NeRF-Oriented 3-D Object Counting for Industrial Fruit Harvesting

Yimo Wang[✉], Bin Kang[✉], *Member, IEEE*, Jian Liu[✉], *Member, IEEE*, and Changyin Sun[✉], *Senior Member, IEEE*

Abstract—Object counting is the core of perception components in the industry system. Nevertheless, real-world occlusion and dense clustering pose significant challenges to fruit harvesting. Existing methods closely rely on physics-agnostic 2D clustering with rigid thresholds, neglecting critical 3D geometric cues in complex scenes. Thus, they suffer from the multi-view double-counting issue. In this paper, we propose a 3D object counting framework that integrates semantic Neural Radiance Fields (NeRF) with a physics-guided adaptive clustering algorithm. In particular, we employ a semantic NeRF to achieve implicit 3D scene reconstruction, which effectively isolates target objects from the background. Based on the semantic NeRF, we introduce a physics-guided adaptive clustering algorithm that exploits physically interpretable features, including surface normals and elevation gradients, for accurate unsupervised point cloud segmentation. Subsequently, an energy function optimization mechanism is utilized to autonomously aggregate multi-dimensional features for dynamically adjusting clustering thresholds, which enables the 3D counting to adapt to diverse fruit morphologies without manual parameter tuning. Extensive experiments on both synthetic and real-world datasets demonstrate the superiority of the proposed framework, which improves counting accuracy by an average of 9.0 percentage points (6.9 pp on synthetic, 7.2 pp on real-world, and 12.9 pp on Fuji) over the state-of-the-art 3-D baseline.

Note to Practitioners—This work targets the challenge of accurate fruit counting in orchards where heavy occlusion and dense clustering often cause double-counting and unreliable yield estimation. Conventional 2D vision systems depend on hand-tuned thresholds and lack spatial awareness. To address this, we propose a 3D counting framework that integrates a semantic Neural Radiance Field (NeRF) with a physics-guided adaptive clustering algorithm. The framework reconstructs the orchard scene in 3D, isolates fruits from the background, and adaptively segments point clouds by analyzing geometric cues such as sur-

face normals and elevation gradients. This approach eliminates the need for manual calibration and improves counting accuracy under varying lighting and viewing conditions. For practitioners, the method can be incorporated into automated yield-estimation or robotic-harvesting systems to enhance reliability and reduce labor costs. The current pipeline targets offline pre-harvest yield estimation within an edge-cloud workflow, where a UAV or ground robot captures imagery in the field and a remote server performs NeRF reconstruction and counting. The current limitation lies in the computational load of NeRF reconstruction. Future work will explore efficient 3D representations such as Gaussian Splatting for real-time deployment in field robots.

Index Terms—Intelligent agriculture, object counting, semantic reconstruction, adaptive point cloud clustering.

I. INTRODUCTION

INDUSTRY 4.0 is expanding into different areas, e.g. Artificial Intelligence (AI), and robotics are increasingly permeating agriculture, giving rise to the paradigm of ‘Agriculture 4.0’. Within this development trend, automated harvesting is a critical component for modernizing orchard management. Specifically, the capability for precise and robust real-time fruit counting and sorting is fundamental for accurate yield prediction and robotic operational planning [1]. This capability is profoundly important. Because accurate in-field counting enables precise yield forecasting essential for optimizing logistics and business strategies, while automated sorting by quality attributes like size and color is vital for minimizing post-harvest losses. Therefore, developing reliable automated counting and sorting technologies is a foundational pillar for the economic viability and efficiency of next-generation smart agriculture systems.

In fruit harvesting contexts, the object counting task is intrinsically complicated by the tendency of dense three-dimensional clusters, resulting in severe occlusion that obscures individual boundaries [4]. Conventional 2D vision-based approaches, such as image processing reliant on color segmentation or edge detection [14], frequently fail to achieve reliable counting in such complex spatial scenarios. Traditional three-dimensional counting methodologies aim to achieve a comprehensive enumeration of target objects by leveraging visual data from images or videos captured across multiple viewpoints. Although deep learning-based 3D counting methods have shown promise in certain scenarios, they often suffer redundant counting issues when aggregating multi-view observations [5], [6], [7], as illustrated in Fig. 1. The core challenge arises from the projective nature of their 2D imaging, wherein different viewpoints capture only partial object

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Yimo Wang is with the School of Automation, Southeast University, Nanjing 210096, China (e-mail: yimowang@seu.edu.cn).

Bin Kang is with the School of Internet of Thing, Nanjing University of Posts and Telecommunications, Nanjing 210003, China (e-mail: kangb@njupt.edu.cn).

Jian Liu is with the School of Automation, Southeast University, Nanjing 210096, China (e-mail: bkliujian@seu.edu.cn).

Changyin Sun is with the School of Automation, Southeast University, Nanjing 210096, China, also with the Key Laboratory of Measurement and Control of Complex System of Engineering, Ministry of Education, Southeast University, Nanjing 210096, China, and also with the School of Artificial Intelligence, Anhui University, Hefei 230601, China (e-mail: cysun@seu.edu.cn).

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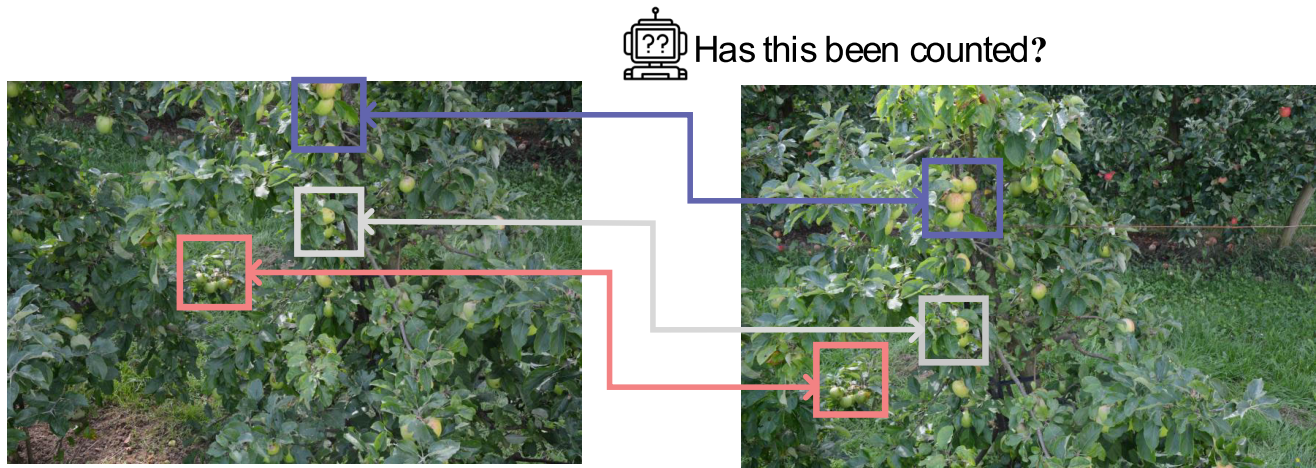


Fig. 1. Double counting issue in multi-view Object Counting. In the context of multi-view image-based counting, variations in viewpoint and the high degree of similarity among target objects make it challenging to disambiguate individual instances across different perspectives, which consequently leads to the problem of erroneous duplicate counting.

appearances, lacking inherent spatial awareness. This results in two primary limitations: on the one hand the inability to distinguish individual object instances in overlapping regions, and on the other hand, the absence of consistent identity tracking across multiple views.

Three-dimensional (3D) reconstruction-based methodologies present a viable avenue for circumventing the inherent limitations of conventional object counting techniques. By generating explicit 3D models from multi-view imagery, these methods furnish a comprehensive spatial context essential for enumeration tasks. Prominent among contemporary 3D reconstruction approaches are Neural Radiance Fields (NeRF) and 3D Gaussian Splatting (3DGS). Effective object counting via 3D reconstruction, however, mandates more than the mere reconstruction of target objects; a crucial prerequisite is the separation of these objects from the background to enable the isolation of their respective point clouds for subsequent enumeration. While 3DGS demonstrates efficiency in scene approximation through the fitting of anisotropic Gaussian kernels, integrating semantic information into Gaussian-splat representations typically requires explicit feature-field distillation that increases pipeline complexity [40]. Conversely, NeRF's architectural flexibility natively permits the incorporation of semantic inputs, such as object masks, into the continuous volumetric field. This facilitates the direct disentanglement of radiance fields pertinent to target objects from convoluted backgrounds. Nevertheless, the direct utilization of point clouds derived from NeRF reconstructions does not comprehensively address the counting problem. This is due to the fact that existing point cloud clustering algorithms, including DBSCAN and Euclidean clustering, are heavily reliant on meticulous hyperparameter tuning and exhibit difficulties in processing objects of irregular morphology [8], [9], [10]. Distance-centric methods often prove inadequate in separating concave surface regions of proximate fruits, whereas density-centric approaches are prone to misclassifying sparsely distributed peripheral points [11]. Furthermore, supervised deep learning methods impose considerable prerequisites [4]. Critically, a majority

of extant approaches neglect vital geometric cues, such as surface normals and elevation differentials, which encapsulate indispensable shape information for clustered fruits [12], [13].

In this paper, we introduce a physics-guided object counting framework that leverages 3D information. Our framework first employs a semantic-aware Neural Radiance Field (NeRF) to isolate target objects from the background. The extracted point cloud is then subjected to a physics-guided adaptive clustering algorithm to perform robust instance segmentation and counting. The primary innovation of our work lies in the physics-guided clustering solution, which presents a new paradigm for instance segmentation in point cloud-based counting tasks. In contrast, conventional clustering methods are often physics-agnostic, only relying on point density or Euclidean distance while ignoring the rich geometric information that encodes physical reality. Consequently, they fail to resolve boundaries between closely-packed or touching fruits where density remains uniformly high, leading to erroneous merging (i.e., under-segmentation) and inaccurate counts.

To model the core physical properties of fruit clusters, the proposed physics-guided clustering method initially employs point density as a primary geometric feature. This feature is utilized to define object contours, as high-density regions typically correspond to an object's body while sparse, low-density areas delineate potential boundaries. However, point density only is insufficient for the fruit counting task due to frequent occlusion and inter-fruit contact. Specifically, in regions of direct physical contact between adjacent fruits, the point density remains uniformly high. This results in erroneously merging distinct instances. To address this critical limitation, our framework introduces two additional, more descriptive geometric features: normal vectors and elevation gradients.

Descriptive geometric features leverage the observation that a surface normal vector's direction remains consistent across a single, continuous object. Consequently, a sharp, abrupt change in vector orientation serves as a powerful physical indicator of a boundary, such as the crease between two touching objects. We capture this trait by analyzing the local

consistency of the normal vectors, where a high variance is treated as strong evidence of an inter-object boundary. In addition to normal vectors, we incorporate elevation. A point's elevation gradient on a fruit's surface is tied to its growth orientation, indicating that a single fruit should exhibit a smooth elevation profile. By evaluating elevation gradients, we can capture the sharp, discontinuous changes that occur at the junction of two separate fruits. This abrupt gradient change is treated as further physical evidence for delineating an instance boundary. Based on the above observation, we operationalize the physical model by integrating density, normal vectors, and elevation gradients within a hierarchical optimization framework. This framework leverages progressive threshold optimization to precisely estimate the point cloud distribution of each fruit in 3D physical space. This adaptive process proficiently distinguishes concave boundary regions between contiguous objects from true topological separations. This endows the 3D object counting framework with the adaptability to accommodate variations in fruit morphology without manual parameter recalibration.

The main contributions of this paper are as follows:

- We propose an object counting framework that formulates the conventional multi-view counting problem into the 3D domain, which fundamentally resolves the duplicate counting issue that arises when aggregating observations from overlapping 2D views.
- We propose a physics-guided adaptive point cloud clustering algorithm in the object counting framework. Our algorithm integrates multi-dimensional geometric features, including surface normals and elevation gradients, into a hierarchical optimization to adapt to diverse fruit morphologies and accurately distinguish between densely clustered or occluded instances without manual parameter tuning.
- Extensive experiments on synthetic and real-world datasets, including Fuji-SfM, validate our method's effectiveness. Our method excels at handling severe occlusion and dense clustering, improving counting accuracy by an average of 9.0 percentage points (6.9 pp on synthetic, 7.2 pp on real-world, and 12.9 pp on Fuji) over the state-of-the-art 3-D baseline.

II. RELATED WORKS

A. Object Counting

Conventional fruit counting methods, relying on manual or semi-automated techniques, are limited in accuracy by occlusion and dense clustering. These approaches are often impeded by dense fruit arrangements [2] and fail to generalize across diverse lighting conditions and fruit morphologies [4]. While 2D deep learning methods have shown potential for handling occlusion in specific applications [14], [15], they are inherently unable to resolve the duplicate counting problem that arises when observing a 3D scene from multiple viewpoints. Although object tracking across image sequences can mitigate this issue [6], a more robust paradigm involves reformulating the problem in the 3D domain. Leveraging modern reconstruction technology, this approach has emerged

as a new research direction. For instance, Meyer et al. utilized a Neural Radiance Field (NeRF) to reconstruct the fruit point cloud, which enables direct counting in 3D space and resolves the issue of inter-frame duplicate counts [3].

B. Neural Radiance Field

Neural Radiance Field (NeRF), introduced by Mildenhall et al., transformed 3D scene representation by using neural networks to implicitly model scenes for high-quality novel view synthesis. This foundational work has since been extended and improved in various aspects [16]. Key advancements include addressing anti-aliasing, with Mip-NeRF introducing conical tracing [17] and Zip-NeRF proposing a grid-based method [18]. Efficiency has also been a central research focus, with methods like Instant-NGP reducing computational costs through adaptive input encoding [19] and TensorRF employing tensor decomposition to accelerate processing [20]. Extending NeRF to semantic modeling is crucial for 3D scene understanding. The initial approach, Semantic-NeRF, integrated a semantic field supervised by annotated labels [21]. However, this method's reliance on extensive annotated data is a significant limitation. To address this, subsequent research has explored unsupervised or self-supervised techniques. For instance, Tschernezki et al. proposed a teacher-student framework [22] that distills features from a pre-trained DINO model [23] into a trainable semantic field.

C. Point Cloud Segmentation

Point cloud segmentation is a fundamental task in 3D analysis, with clustering methods often categorized by their underlying principles. Density-based methods, such as DBSCAN [10], define clusters based on local point density, which is effective for identifying irregularly shaped groups. However, their efficacy is limited by a sensitivity to manually tuned parameters (e.g., neighborhood radius) and outliers, which can disrupt density connectivity, a notable challenge in fruit counting applications. Distance-based methods, including k-means [24] and KD-tree Euclidean clustering [25], [26], rely on fixed distance thresholds or initial centers. This reliance makes them susceptible to errors in complex scenes; k-means is sensitive to initialization and may mispartition non-spherical fruits, while Euclidean clustering's uniform thresholds perform poorly on heterogeneous data. Region-growing methods [27] expand clusters based on local similarity metrics. While effective for coherent surfaces, their strict connectivity requirements can lead to over-segmentation of sparse point clouds, such as those representing individual fruits. Model-based approaches [13] fit geometric primitives (e.g., spheres, planes) to the data. This is efficient for regular structures but fails for irregular objects like fruits, where the limited representational power of primitives leads to poor segmentation. Graph-based methods [28], [29] excel at encoding complex spatial relationships but often incur high computational costs, limiting their applicability to large-scale point clouds or real-time scenarios.

III. METHOD

A. Pipeline Overview

The proposed framework processes multi-view RGB images in three sequential stages. First, a Structure-from-Motion (SfM) pipeline recovers camera parameters. Second, the posed images and fruit segmentation masks jointly supervise a semantic NeRF, from which a fruit-only point cloud is extracted via orthographic ray-marching. Third, the point cloud is fed to our physics-guided adaptive clustering algorithm, which identifies individual fruit instances and produces the final count. Within this pipeline, the SfM module, the Nerfacto backbone, and the segmentation priors (Grouned-SAM or U-Net) are adopted without modification; the *instance-decoupled semantic NeRF formulation* (Section III-C) and the *physics-guided adaptive clustering algorithm* (Section III-E) constitute our novel contributions. The framework assumes that each fruit instance is approximately convex and at least partially visible from the camera trajectory, so that its surface can be reconstructed with sufficient point density for reliable normal estimation.

B. Structure-From-Motion Preliminaries

Training a Neural Radiance Field (NeRF) requires robust camera poses and intrinsics, which are typically obtained via a Structure-from-Motion (SfM) pipeline. Given an unordered image set $\mathcal{I} = \{I_1, \dots, I_N\}$, the initial step is to establish geometric correspondences by matching features across views. For a feature i in image I_k with 2D coordinates $\mathbf{u}_i^{(k)} \in \mathbb{R}^2$ and descriptor $\mathbf{d}_i^{(k)}$, the matching process identifies pairs between images (I_p, I_q) that satisfy $\|\mathbf{d}_i^{(p)} - \mathbf{d}_j^{(q)}\|_2 < \epsilon$, where ϵ is a similarity threshold.

Once matches are established, the camera projection is modeled. Each camera k is defined by an intrinsic matrix $\mathbf{K}_k$ and extrinsic parameters (rotation $\mathbf{R}_k \in SO(3)$ and translation $\mathbf{t}_k \in \mathbb{R}^3$). The perspective projection from a 3D world point $\mathbf{X} = [X, Y, Z]^T$ to its 2D image projection $\mathbf{u} = [u, v]^T$ is given by:

$$\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \mathbf{K}_k [\mathbf{R}_k | \mathbf{t}_k] \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix} \quad (1)$$

where λ is a depth scale factor. The intrinsic matrix is parameterized as $\mathbf{K}_k = \begin{bmatrix} f_k & 0 & c_x \\ 0 & f_k & c_y \\ 0 & 0 & 1 \end{bmatrix}$, assuming a shared focal length f_k .

Finally, the SfM pipeline refines all camera parameters and the 3D scene structure by solving a large-scale non-linear optimization known as Bundle Adjustment (BA). This process minimizes the total reprojection error—the discrepancy between projected 3D points and their observed 2D feature locations. The objective function is:

$$\arg \min_{\{\mathbf{K}_k, \mathbf{R}_k, \mathbf{t}_k\}, \{\mathbf{X}_j\}} \sum_{k=1}^N \sum_{j \in \mathcal{V}_k} \rho \left(\left\| \pi(\mathbf{K}_k, \mathbf{R}_k, \mathbf{t}_k, \mathbf{X}_j) - \mathbf{u}_j^{(k)} \right\|_2^2 \right) \quad (2)$$

Here, $\mathcal{V}_k$ is the set of 3D points visible in image I_k , $\pi(\cdot)$ is the projection function, and $\rho(\cdot)$ is a robust cost function

(e.g., Cauchy loss) to reduce the influence of outliers. This optimization yields a set of refined camera parameters and a sparse 3D point cloud, providing the essential geometric priors for subsequent NeRF training.

C. Instance-Decoupled Reconstruction With Semantic Guidance

Our framework employs semantic-aware neural reconstruction to achieve instance-level decomposition, enabling precise segregation of target objects from cluttered backgrounds. As illustrated in Fig. 2, our approach extends the classical NeRF by jointly modeling scene appearance and semantics. The model comprises three core components:

- **Density Field:** $\mathcal{F}_\sigma : \mathbf{x} \rightarrow \sigma$, maps a 3D coordinate $\mathbf{x} \in \mathbb{R}^3$ to a volume density $\sigma \in \mathbb{R}$.
- **Appearance Field:** $\mathcal{F}_c : (\mathbf{x}, \mathbf{d}) \rightarrow \mathbf{c}$, maps a coordinate $\mathbf{x}$ and viewing direction $\mathbf{d} \in \mathbb{S}^2$ to an RGB color $\mathbf{c} \in \mathbb{R}^3$.
- **Semantic Field:** $\mathcal{F}_s : \mathbf{x} \rightarrow s$, maps a coordinate $\mathbf{x}$ to a semantic logit $s \in \mathbb{R}$ to distinguish target from background. Crucially, $\mathcal{F}_s$ is conditioned only on position and the gradient of the semantic loss is back-propagated solely through $\mathcal{F}_s$, so the learned fruit occupancy is invariant to view-dependent illumination effects.

1) *Volume Rendering:* For a camera ray $\mathbf{r}(t) = \mathbf{o} + t\mathbf{d}$, we sample K points along the ray to compute the pixel's color and semantic values. The color $\hat{\mathbf{C}}(\mathbf{r})$ is rendered via the equation:

$$\hat{\mathbf{C}}(\mathbf{r}) = \sum_{k=1}^K T(t_k) \alpha_v(\sigma(t_k) \delta_k) \mathbf{c}(t_k) \quad (3)$$

where $T(t_k) = \exp\left(-\sum_{a=1}^{k-1} \sigma(t_a) \delta_a\right)$ is the accumulated transmittance, $\alpha_v(x) = 1 - \exp(-x)$ is the absorption probability, and δ_k is the distance between adjacent samples.

Similarly, the semantic logit $\hat{s}(\mathbf{r})$ is rendered as:

$$\hat{s}(\mathbf{r}) = \sum_{k=1}^K T(t_k) \alpha_v(\sigma(t_k) \delta_k) s(t_k) \quad (4)$$

where $s(t_k)$ is the semantic logit at the sample point. The corresponding semantic probability is obtained via the sigmoid activation $\hat{p}(\mathbf{r}) = \sigma_g(\hat{s}(\mathbf{r})) \in [0, 1]$, where $\sigma_g(\cdot)$ denotes the logistic sigmoid (distinguished from the density σ).

D. Point Cloud Extraction From the Reconstructed Volume

For subsequent analysis, we export the semantically reconstructed scene as a point cloud. We use a ray-marching technique with an orthographic camera setup to sample the continuous volumetric representation. For each ray $\mathbf{r}(t) = \mathbf{o} + t\mathbf{d}$, we sample points $\mathbf{x}_k$ at fixed intervals.

At each sample point $\mathbf{x}_k$, we query the density field to obtain $\sigma_k = \mathcal{F}_\sigma(\mathbf{x}_k)$ and the semantic probability $\hat{p}_k = \sigma_g(\mathcal{F}_s(\mathbf{x}_k))$. A point is retained if its density exceeds the density threshold τ_σ (indicating proximity to a surface) and its semantic probability exceeds the semantic threshold τ_s (indicating fruit class). The final point cloud $\mathcal{P}$ is defined as:

$$\mathcal{P} = \{\mathbf{x}_k \mid \sigma_k > \tau_\sigma \wedge \hat{p}_k > \tau_s\} \quad (5)$$

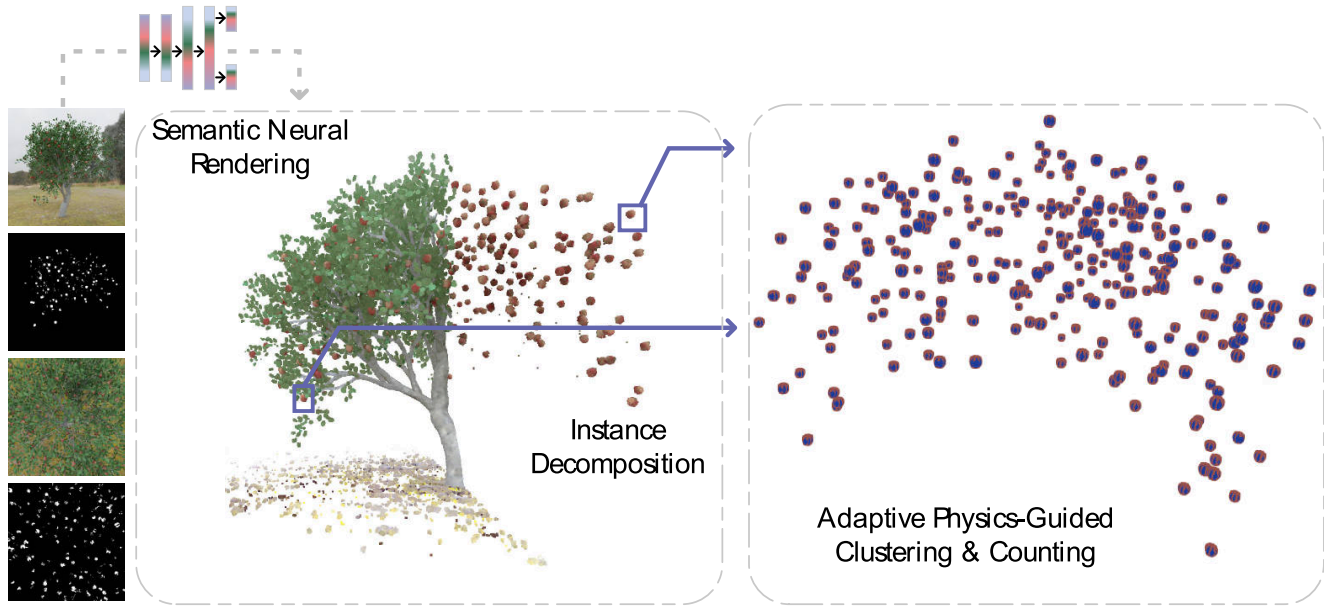


Fig. 2. Overview of the proposed counting framework. The pipeline comprises three sequential stages: SfM-based pose recovery, semantic neural rendering (producing a fruit-only point cloud $\mathcal{P}$), and physics-guided adaptive clustering (producing the fruit count). Coloured points in the clustering stage indicate distinct fruit instances.

To ensure comprehensive coverage, points sampled from all six orthographic views are aggregated. This multi-view sampling enhances the point cloud's density and completeness, mitigating occlusions present in any single perspective.

E. Adaptive Physics-Guided Point Cloud Segmentation

We propose an adaptive, shape-based point cloud clustering method to accurately estimate fruit counts from the reconstructed point cloud. Our approach is distinguished by two key innovations. First, it employs **shape-based clustering**, which leverages geometric information such as surface normals rather than relying solely on Euclidean distance or point density, thereby capturing the morphological characteristics of fruits more accurately. Second, its **adaptive nature** eliminates the need to manually specify the number of clusters or density thresholds, enhancing robustness to initial conditions via automated representative selection and dynamic thresholding. The proposed method consists of three main steps: point cloud preprocessing, adaptive clustering, and counting.

1) *Point Cloud Preprocessing*: Given an input fruit point cloud $\mathcal{P} = \{p_j\}_{j=1}^M$ extracted from the 3D reconstruction, where $p_j = (x_j, y_j, z_j)$, we first preprocess the data to improve the efficiency and quality of subsequent clustering. The process begins with downsampling, where a voxel grid filter is applied to reduce the point cloud resolution to a uniform grid size v , yielding a downsampled point cloud $\mathcal{P}' = \{p'_k\}_{k=1}^{M'}$ and reducing computational complexity. This step also enforces a uniform spatial resolution across fruit types of different sizes, which makes the dynamic threshold below scale-invariant. Following this, a statistical outlier removal filter is employed to eliminate noise points based on an outlier-distance threshold

τ_{out} derived from the average distance to neighbouring points, producing a clean point cloud for clustering.

2) *Adaptive Clustering*: Adaptive clustering forms the core of our method, designed to identify individual fruit instances from the preprocessed point cloud $\mathcal{P}'$ without manual intervention. This is achieved through automatic representative selection and shape-based dynamic threshold clustering.

a) *Automatic representative selection*: Conventional clustering approaches often rely on predefined parameters that do not adapt well to variable fruit morphologies. To overcome this, we formulate an adaptive framework based on energy function optimization to automatically determine optimal cluster representatives while maintaining spatial consistency. The framework operates without prior assumptions about the number of clusters, as defined by the following objective function:

$$E = \sum_{i=1}^{M'} \min_{j=1}^K d(p'_i, r_j) + \lambda K \quad (6)$$

where $p'_i \in \mathcal{P}'$ is a point, r_j is the j -th representative, $d(\cdot, \cdot)$ is the Euclidean distance, and λ is a regularization parameter controlling the number of representatives. The objective is to minimize E , balancing the distance from points to their nearest representatives against the total number of representatives. We solve this using a bottom-up merging strategy. The algorithm is initialized by setting the representative set to be the entire point cloud, $R = \mathcal{P}'$, and initializing λ as the median of nearest neighbor distances. In each iteration, for every representative r_i , its nearest neighbor r_j is identified, and the energy change $\Delta E = \lambda - (\text{points managed by } r_j) \times d(r_i, r_j)$ upon merging is computed. If $\Delta E > 0$, the representatives are merged, and the set R is updated. After each iteration, λ is

doubled to progressively reduce the number of representatives. The doubling schedule $\lambda \leftarrow 2\lambda$ is chosen because the optimal cluster count scales approximately as $K^* \propto 1/\lambda$; doubling therefore halves the expected number of remaining representatives at each step and guarantees convergence in $O(\log M')$ iterations. The process continues until $|\Delta E| < 10^{-4}$, yielding the final representative set $R = \{r_j\}_{j=1}^K$.

b) *Shape-based dynamic threshold clustering*: Using the final representative set R , we employ a depth-first search (DFS) to cluster the point cloud, assigning each point p'_i to its nearest representative. To enhance accuracy, we incorporate surface normal information and a dynamic threshold. The normal $\mathbf{n}_i$ of each point is estimated via principal component analysis (PCA) of its local neighborhood. The dynamic threshold, θ , is defined as:

$$\theta = \alpha \cdot \delta \cdot \frac{\|p'_i - p'_j\|}{|p'_{i,z} - p'_{j,z}| \cdot \arccos(\kappa(\mathbf{n}_i \cdot \mathbf{n}_j))} \quad (7)$$

This threshold integrates three complementary geometric cues that together provide a sufficient description of a fruit boundary. The *normal-consistency* term $\arccos(\kappa(\mathbf{n}_i \cdot \mathbf{n}_j))$ detects the sharp angular discontinuity that appears at the crease between two touching convex fruits, which is the dominant signal when fruits are in direct contact and distance-based cues fail. The *elevation-gradient* term $|p'_{i,z} - p'_{j,z}|/\|p'_i - p'_j\|$ captures the height-profile discontinuity of vertically stacked fruits, and is complementary to the normal cue where adjacency is near-tangential. The *local density* δ weights both cues by the reconstruction reliability of the local region, tightening the threshold in sparse or occluded areas and relaxing it in well-sampled ones. Because horizontal and vertical contacts are the two principal contact modes in gravity-aligned orchard scenes, these cues provide sufficient coverage of the boundary geometries encountered in practice. Here, α is a scalar adjustment factor that scales the composite score to the Euclidean distance units used by the DFS expansion, and $\kappa(\cdot)$ is a clipping function that confines the dot product to the valid input range of $\arccos$ to avoid floating-point-induced numerical failure. The clustering process is detailed in Algorithm 1.

3) *Fruit Counting*: Finally, the fruit count is estimated from the optimised clusters $\{C'_j\}_{j=1}^K$. Each cluster C'_j is treated as a potential fruit instance. We compute the convex-hull volume of each cluster:

$$V_j = \text{ConvexHull}(C'_j). \text{volume} \quad (8)$$

No externally specified reference volume is required: we derive the reference directly from the clustering output as the median of the per-cluster volumes,

$$V_{\text{ref}} = \text{median}(\{V_j\}_{j=1}^K), \quad (9)$$

which is robust to under- or over-segmented outliers since the median has a 50% breakdown point. The per-cluster fruit count

Algorithm 1 Adaptive Shape-Based Clustering for Fruit Counting

Input: Preprocessed point cloud $\mathcal{P}' = \{p'_k\}_{k=1}^{M'}$, regularization parameter λ , adjustment factor α

Output: Fruit count N

Initialization: Set representative set $R = \mathcal{P}'$, initialize λ as the median of nearest neighbor distances

Step 1: Automatic Representative Selection

while energy E not converged ($|\Delta E| > 10^{-4}$) **do**

 Update representatives R by merging based on energy change ΔE

 Adjust $\lambda \leftarrow 2 \times \lambda$

end while

Obtain final representatives $R = \{r_j\}_{j=1}^K$

Step 2: Adaptive Clustering with Shape Features

for each point p'_i in $\mathcal{P}'$ **do**

 Compute normal $\mathbf{n}_i$ using PCA

 Assign p'_i to nearest representative r_j with dynamic threshold θ from Eq. (7)

 Expand cluster C_j using DFS

end for

Step 3: Cluster Boundary Optimization

while clusters not converged **do**

 Reassign points to dominant cluster labels in their neighborhood

end while

Obtain optimized clusters $\{C'_j\}_{j=1}^K$

Step 4: Fruit Counting

Compute V_j for each cluster via Eq. (8); compute V_{ref} via Eq. (9)

for each cluster C'_j **do**

 Estimate count $_j$ via Eq. (10)

end for

Compute total fruit count $N = \sum_{j=1}^K \text{count}_j$

is then assigned by comparing V_j to V_{ref} within a tolerance band:

$$\text{count}_j = \begin{cases} 1, & \text{if } 0.9V_{\text{ref}} \leq V_j \leq 1.1V_{\text{ref}}, \\ \lfloor V_j/V_{\text{ref}} \rfloor, & \text{if } V_j > 1.1V_{\text{ref}}, \\ 0, & \text{if } V_j < 0.9V_{\text{ref}}. \end{cases} \quad (10)$$

The first case handles single-fruit clusters; the second handles multi-fruit (under-segmented) clusters, exploiting that the convex-hull volume of n touching convex fruits scales approximately linearly in n ; the third filters out noise fragments. The total count is $N = \sum_{j=1}^K \text{count}_j$. This volume-based rule is parameter-free and adapts across fruit types (because V_{ref} is recomputed per scene), in contrast to template-matching approaches that require a predefined exemplar.

The overall complexity of the clustering pipeline is $O(M' \log^2 M')$, dominated by the energy-minimisation step, which is near-linear in the point count and scales gracefully to large orchard scenes.

IV. EXPERIMENT

This section details the experimental evaluation of our proposed object counting framework. We first outline the datasets

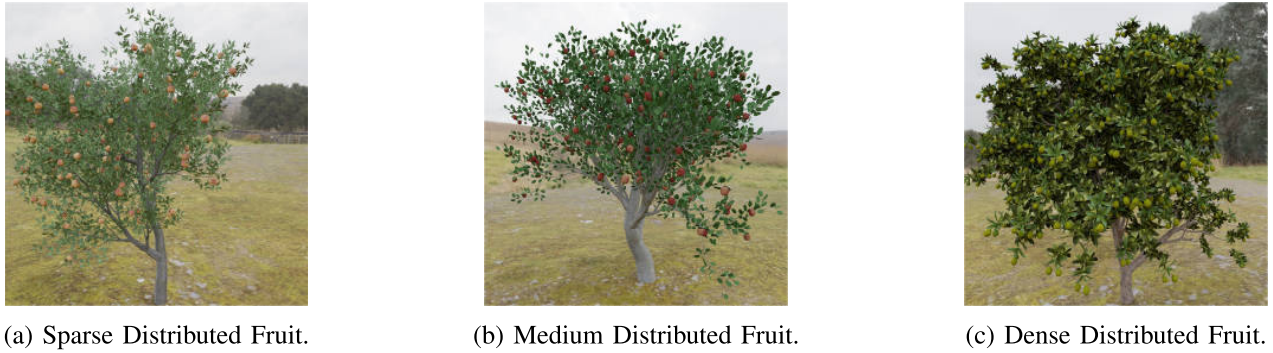


Fig. 3. Visualisation of the synthetic dataset with three controlled fruit-density levels.

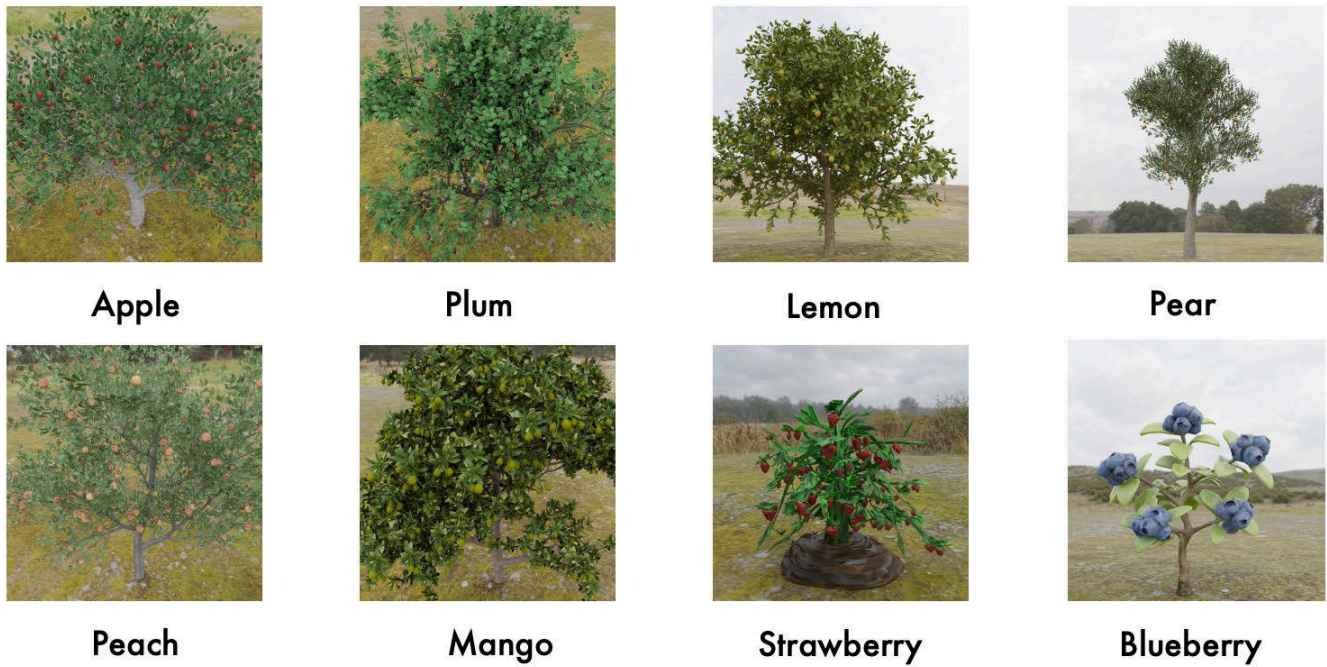


Fig. 4. Rendered example images for all eight fruit types in the synthetic dataset: apple, plum, lemon, pear, peach, mango, strawberry, and blueberry.

and experimental setup, followed by a comparative analysis against state-of-the-art methods. We then provide a qualitative analysis through visualisations, investigate the impact of various factors via ablation, and report the computational cost.

A. Datasets and Evaluation Protocol

For our evaluation, we utilise publicly available synthetic and real-world datasets from FruitNeRF [3], supplemented by the Fuji-SfM dataset [31], to ensure objectivity and comparability. We report Accuracy (Acc), Recall, F1-score, Mean Absolute Error (MAE), and Root Mean Squared Error (RMSE). Here, $\text{Acc} = 1 - |N_{\text{pred}} - N_{\text{gt}}|/N_{\text{gt}}$ denotes *scene-level counting accuracy*, whereas Recall and F1-score are computed from instance-level fruit matching. For the 2D baselines in Table II, the first-column values follow the precision numbers reported in the original papers because scene-level count accuracy is not available. The same hyper-parameter suite is used across all datasets without per-dataset tuning.

1) Synthetic Dataset: The synthetic dataset, generated using Blender with the BlenderNeRF plugin [30], comprises eight fruit types: apple, plum, lemon, pear, peach, mango, strawberry, and blueberry. Each tree is represented by 300 images at a resolution of 1024×1024 px. To enable systematic performance assessment, the dataset features controlled variations across two dimensions: occlusion level and fruit density, categorised into three levels based on inter-fruit spacing: **sparse** (> 5 cm), **medium** (2–5 cm), and **dense** (< 2 cm), as visualised in Figure 3. Ground-truth counts are read directly from the scene graph, eliminating annotation uncertainty. Rendered examples for all eight fruit types are shown in Figure 4. Note that the strawberry and blueberry scenes are modelled with sufficient spatial separation that each berry is at least partially visible from the camera hemisphere; the most extreme natural clustering configurations of these two species, in which inner berries are permanently occluded, fall outside the applicability scope of any multi-view reconstruction pipeline.

TABLE I
SOFTWARE LIBRARIES AND TOOLS USED IN THE MAIN PIPELINE STAGES

Stage	Library / Tool	Version
Camera pose recovery	COLMAP [41]	3.8
Semantic segmentation (SAM)	Grounded-SAM [32]	v1.1
Semantic segmentation (U-Net)	PyTorch U-Net [33]	custom
NeRF training & export	Nerfstudio (Nerfacto) [42]	0.3.4
Point-cloud processing	Open3D [43]	0.17.0
Clustering & convex hull	NumPy / SciPy	1.24 / 1.11
Visualisation	Open3D / Matplotlib	0.17.0 / 3.8

2) *Real-World Dataset*: The real-world evaluation uses the three apple-tree scenes (Tree 01–03) from the FruitNeRF benchmark (approx. 350 images per tree at 1000×1500 px, captured with a Nikon D7100 at 35 mm focal length, with per-tree ground-truth counts 179/113/291 obtained by on-site manual counting), together with the Fuji-SfM dataset [31], which contains 11 Fuji apple trees in a single row with 582 images and per-fruit 3D annotations. We adopt both datasets and their published ground truth without modification.

B. Experiment Setup

All experiments are conducted on an NVIDIA RTX 3090 GPU with an AMD Ryzen 9 5950X CPU, running Ubuntu 22.04 with CUDA 11.8. The main software components are summarised in Table I. Unless otherwise stated, the fixed implementation constants are: orthographic ray step $\Delta s = 0.004$ (normalised units), density threshold $\tau_\sigma = 0.01$, semantic threshold $\tau_s = 0.5$, voxel size $v = 0.01$ m, statistical-outlier $k = 20$, outlier multiplier $\sigma_{th} = 2.0$, normal-estimation radius $r_n = 0.05$ m with $k_n = 8$ neighbours, adjustment factor $\alpha = 0.5$, λ_0 initialised as the median nearest-neighbour distance and doubled each iteration, and $\kappa(x) = \max(-1 + \varepsilon, \min(1 - \varepsilon, x))$ with $\varepsilon = 10^{-6}$.

We evaluate our method against 2D and 3D baselines on all datasets. To analyse the effect of the upstream semantic segmentation and reconstruction, we compare two segmentation strategies—a pre-trained Grounded-SAM model [32] prompted with “fruit” and a U-Net instance-segmentation model [33] trained on annotated data—and two NeRF capacities: a **Small** model (2 hidden layers, 64 units) and a **Large** model (3 layers, 128 units).

Model training is guided by a joint loss function comprising a photometric loss and a semantic loss. The photometric loss measures the mean squared error (MSE) between rendered and ground-truth pixel colors:

$$\mathcal{L}_{\text{Photo}} = \frac{1}{|\mathcal{R}|} \sum_{\mathbf{r} \in \mathcal{R}} \|\mathbf{C}(\mathbf{r}) - \hat{\mathbf{C}}(\mathbf{r})\|_2^2 \quad (11)$$

where $\mathcal{R}$ is the set of sampled rays, and $\mathbf{C}(\mathbf{r})$ and $\hat{\mathbf{C}}(\mathbf{r})$ are the ground-truth and predicted colors, respectively.

The semantic loss employs binary cross-entropy (BCE) to optimize the classification between target (fruit) and background:

$$\mathcal{L}_{\text{Sem}} = -\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r} \in \mathcal{R}} [p(\mathbf{r}) \log \hat{p}(\mathbf{r}) + (1 - p(\mathbf{r})) \log(1 - \hat{p}(\mathbf{r}))] \quad (12)$$

where $p(\mathbf{r}) \in \{0, 1\}$ is the ground-truth label, and $\hat{p}(\mathbf{r}) = \sigma_g(\hat{S}(\mathbf{r}))$ is the predicted probability. The total loss is a sum of these two components:

$$\mathcal{L} = \mathcal{L}_{\text{Photo}} + \mathcal{L}_{\text{Sem}} \quad (13)$$

Crucially, the gradient of $\mathcal{L}_{\text{Sem}}$ is back-propagated only through $\mathcal{F}_s$ and not through the shared Density Field, preventing the density estimate from collapsing to fruit-only surfaces.

C. Experiment Results

1) *Quantitative Results*: To validate the efficacy of our proposed method, we conduct extensive experiments on three distinct datasets: a synthetic dataset, a real-world dataset, and the Fuji dataset. Our approach is benchmarked against a comprehensive suite of baseline methods, encompassing density-based techniques (GMN [34], FamNet [35], CounTR [36]), detection-based approaches (FR [37], FSOD [38], PseCo [39]), and the state-of-the-art 3D counting method, FruitNeRF [3]. Given the scarcity of methods for fruit counting in the 3D domain, we selected the most representative and advanced model, FruitNeRF, for comparison. Table II presents the quantitative comparison across all datasets using the five standard evaluation metrics: Acc, Recall, F1-score, MAE, and RMSE.

On the synthetic dataset, 2D methods exhibit varied performance. CounTR demonstrates robust performance with an F1-score of 0.818 and an MAE of 18.74. In contrast, FR records a markedly lower F1-score of 0.570 and a substantial MAE of 44.17, reflecting inferior performance, while PseCo achieves a moderate F1-score of 0.700. This trend persists in real-world datasets, where CounTR maintains high efficacy (F1 0.833, MAE 19.27), whereas GMN and FSOD display larger errors, with MAEs of 31.39 and 35.76, respectively. On the challenging Fuji dataset, which emulates complex agricultural scenarios, CounTR’s F1-score declines to 0.721, indicating performance limitations under intricate conditions. Although these methods perform adeptly in single-view scenarios, their efficacy diminishes in complex three-dimensional environments. Lacking an understanding of spatial distributions, these methods are unable to effectively distinguish the same target across different viewpoints, which results in redundant counting.

In contrast, FruitNeRF significantly enhances counting accuracy by reformulating the problem in a three-dimensional space, achieving F1 = 0.750 / MAE = 15.75 / RMSE = 31.42 on synthetic, F1 = 0.783 / MAE = 16.94 / RMSE = 25.87 on real-world, and F1 = 0.760 / MAE = 20.86 / RMSE = 47.36 on Fuji. However, its performance is constrained by its reliance on the DBSCAN clustering algorithm and predefined object templates: the initialisation process significantly impacts clustering performance, and the morphological variability of real-world fruits makes precise counting with predefined templates difficult.

Our proposed method builds upon the 3D counting paradigm by integrating adaptive point-cloud clustering with the fusion of geometric features, thereby enhancing both accuracy and robustness. We achieve Acc:0.996, F1:0.977,

TABLE II

QUANTITATIVE COMPARISONS ON THE THREE DATASETS. THE FIRST AND SECOND PARTS CONTAIN THE DENSITY-BASED AND DETECTION-BASED METHODS. THE THIRD PART FOCUSES ON THE STATE-OF-THE-ART COUNTING METHODS IN THREE-DIMENSIONAL SCENES. THE BEST RESULTS ARE SHOWN IN BOLD. HERE, ACC DENOTES SCENE-LEVEL COUNTING ACCURACY $1 - |N_{\text{pred}} - N_{\text{gt}}|/N_{\text{gt}}$ FOR THE 3D METHODS; FOR THE 2D BASELINES, THE FIRST-COLUMN VALUES ARE THE PRECISION NUMBERS REPORTED IN THEIR ORIGINAL PAPERS

Backbone	Synthetic Dataset					Real-World Dataset					Fuji Dataset				
	Acc	Recall	F1-score	MAE	RMSE	Acc	Recall	F1-score	MAE	RMSE	Acc	Recall	F1-score	MAE	RMSE
GMN [34]	0.755	0.715	0.734	29.27	126.17	0.774	0.733	0.752	31.39	116.87	0.651	0.619	0.635	36.67	163.92
FamNet [35]	0.79	0.753	0.77	23.93	101.89	0.808	0.768	0.787	21.75	105.63	0.692	0.661	0.676	30.06	107.05
CounTR [36]	0.837	0.801	0.818	18.74	98.51	0.855	0.812	0.833	19.27	86.80	0.736	0.708	0.721	23.87	119.79
FR [37]	0.586	0.557	0.57	44.17	144.18	0.616	0.58	0.597	39.49	149.45	0.469	0.448	0.458	63.45	171.08
FSOD [38]	0.658	0.633	0.645	34.13	143.96	0.678	0.646	0.661	35.76	129.87	0.541	0.519	0.529	45.65	185.56
PseCo [39]	0.718	0.684	0.7	19.62	125.49	0.737	0.701	0.718	18.15	134.45	0.612	0.587	0.599	24.63	135.42
FruitNeRF [3]	0.927	0.74	0.75	15.75	31.42	0.907	0.689	0.783	16.94	25.87	0.787	0.64	0.76	20.86	47.36
Ours	0.996	0.955	0.977	13.26	25.38	0.979	0.942	0.960	14.15	19.33	0.916	0.970	0.840	16.14	34.84

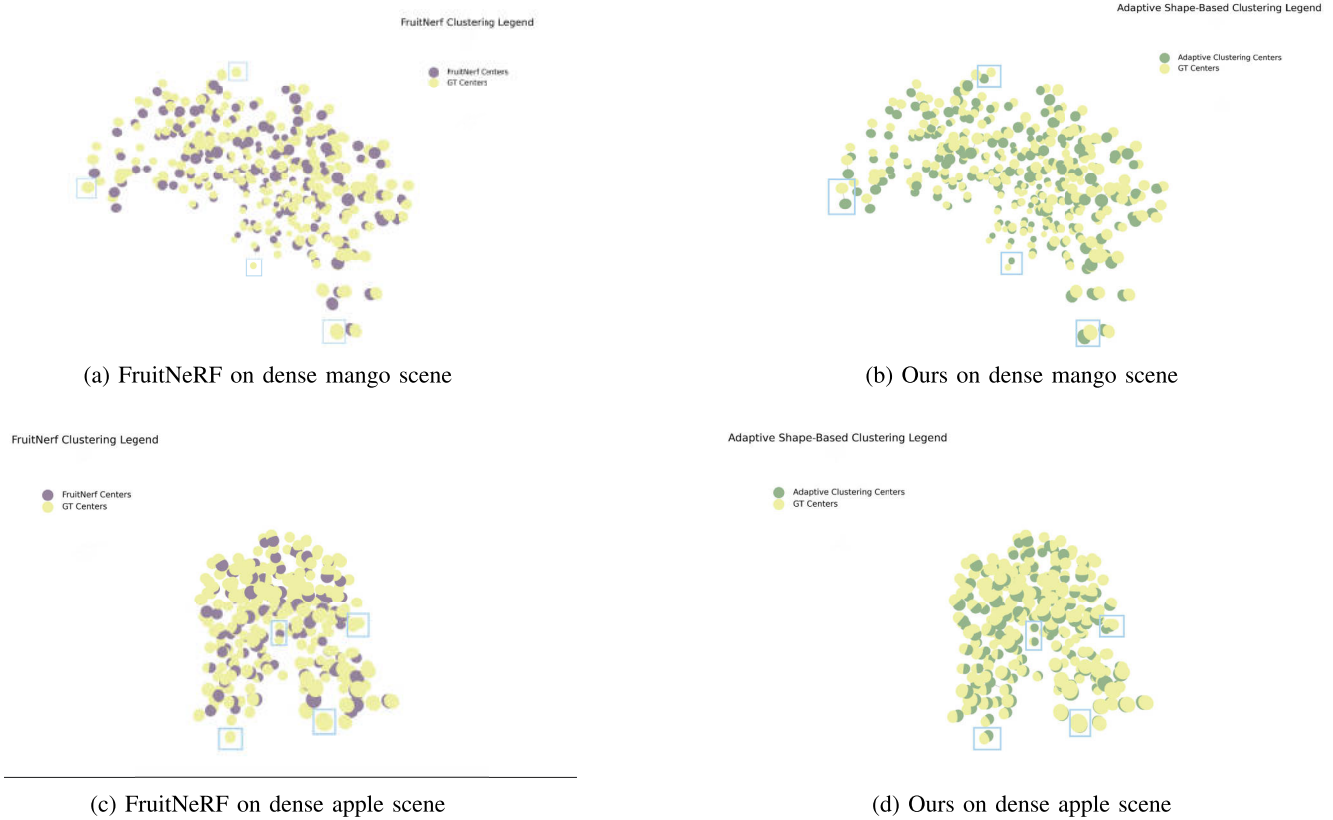


Fig. 5. Qualitative counting results for 3D methods on the dense mango and apple scenes. Coloured points denote per-cluster assignments. Our method correctly separates touching fruits at the canopy apex and base, where FruitNeRF exhibits missed detections and merged clusters.

MAE:13.26, RMSE:25.38 on synthetic, Acc:0.979, F1:0.960, MAE:14.15, RMSE:19.33 on real-world, and Acc:0.916, F1:0.840, MAE:16.14, RMSE:34.84 on Fuji, consistently outperforming FruitNeRF across all datasets. These results demonstrate stronger robustness to variation in fruit type, morphology, and scene complexity, as further validated by the statistical significance test below.

Statistical significance. To assess whether the MAE improvements over FruitNeRF are statistically significant, we performed a paired Wilcoxon signed-rank test [44] on per-tree MAE values from the real-world apple dataset (Trees 01–03) and the Fuji-SfM dataset (11 trees), totalling $n = 14$ paired observations. All 14 per-tree MAE differences

favour our method, yielding the maximum possible test statistic $W^+ = 105$ and a two-sided p -value $p < 0.001$, well below the $\alpha = 0.01$ significance level.

Accuracy improvement. The accuracy gains over FruitNeRF, expressed as absolute differences in Acc (percentage points), are 6.9 pp on synthetic, 7.2 pp on real-world, and 12.9 pp on Fuji.

2) *Qualitative Results:* Figure 5 illustrates the qualitative results of the 3D counting methods. While 2D approaches such as CounTR can manage counting tasks from a single perspective, they inadequately address the issue of redundant counting inherent in 3D scenes. Our comparative analysis therefore focuses on the 3D method, FruitNeRF.

TABLE III
RECONSTRUCTION QUALITY OF SEMANTIC NeRF WITH DIFFERENT SEGMENTATION METHODS ON PEAR, MANGO, AND LEMON DATASETS

NeRF Model	Segmentation Method	Pear				Mango				Lemon			
		PSNR	SSIM	LPIPS	IoU	PSNR	SSIM	LPIPS	IoU	PSNR	SSIM	LPIPS	IoU
Large NeRF	SAM	22.41	0.53	0.45	0.43	21.32	0.52	0.48	0.52	21.08	0.51	0.47	0.53
Large NeRF	U-Net	22.65	0.56	0.43	0.32	21.42	0.52	0.48	0.60	21.47	0.53	0.46	0.59
Small NeRF	SAM	22.83	0.54	0.42	0.46	20.75	0.49	0.45	0.49	20.91	0.50	0.46	0.51
Small NeRF	U-Net	22.91	0.54	0.41	0.38	20.92	0.50	0.44	0.64	21.35	0.51	0.48	0.57

TABLE IV
QUANTITATIVE COMPARISONS OF COUNTING PERFORMANCE ON PEAR, MANGO, AND LEMON DATASETS. HERE, ACC DENOTES SCENE-LEVEL COUNTING ACCURACY $1 - |N_{\text{pred}} - N_{\text{gt}}|/N_{\text{gt}}$

Backbone	Pear				Mango				Lemon			
	Acc	Recall	F1	MAE	Acc	Recall	F1	MAE	Acc	Recall	F1	MAE
Large NeRF + SAM	0.92	0.88	0.90	16.5	0.91	0.85	0.88	18.2	0.89	0.87	0.88	17.1
Large NeRF + U-Net	0.97	0.95	0.96	12.3	0.96	0.94	0.95	13.0	0.95	0.92	0.94	14.1
Small NeRF + SAM	0.93	0.89	0.91	15.4	0.90	0.87	0.88	17.6	0.88	0.86	0.87	18.3
Small NeRF + U-Net	0.95	0.93	0.94	13.2	0.94	0.92	0.93	14.0	0.93	0.90	0.92	15.2

TABLE V
ABLATION STUDY ON THE REGULARIZATION PARAMETER λ . THIS TABLE SHOWS THE SENSITIVITY OF THE MODEL'S PERFORMANCE METRICS (ACC, RECALL, F1-SCORE, AND MAE) TO VARIATIONS IN THE INITIAL λ VALUE, SCALED BY A FACTOR β . HERE, ACC DENOTES SCENE-LEVEL COUNTING ACCURACY $1 - |N_{\text{pred}} - N_{\text{gt}}|/N_{\text{gt}}$. THE BEST RESULT IN EACH COLUMN IS SHOWN IN BOLD

β (Factor)	Synthetic Dataset				Real-World Dataset				Fuji Dataset			
	Acc	Recall	F1-score	MAE	Acc	Recall	F1-score	MAE	Acc	Recall	F1-score	MAE
0.10	0.721	0.995	0.837	81.40	0.705	0.980	0.820	75.30	0.560	0.990	0.715	95.30
0.25	0.865	0.990	0.923	43.00	0.850	0.970	0.906	40.10	0.730	0.985	0.839	48.10
0.50	0.950	0.980	0.965	21.70	0.940	0.950	0.945	22.00	0.850	0.980	0.910	24.50
0.80	0.985	0.965	0.975	14.50	0.975	0.930	0.952	15.30	0.890	0.975	0.931	18.00
1.00	0.996	0.955	0.977	13.26	0.979	0.942	0.960	14.15	0.916	0.970	0.840	16.14
1.20	0.998	0.930	0.963	18.10	0.985	0.880	0.929	19.50	0.925	0.850	0.886	19.30
2.00	0.998	0.860	0.924	34.50	0.990	0.800	0.885	44.20	0.940	0.650	0.769	49.00
5.00	0.999	0.640	0.780	90.10	0.992	0.550	0.708	105.00	0.950	0.400	0.563	112.40
10.00	0.999	0.450	0.621	125.80	0.993	0.360	0.529	145.10	0.960	0.250	0.397	155.00

As depicted in Figs. 5(b) and 5(d), our method achieves more precise detection of fruits under high-density distributions, yielding counting results that more closely align with the ground truth. Moreover, in comparison with FruitNeRF, our approach exhibits a lower rate of missed detections. Even at the apex and base of the fruit trees, where enumeration is difficult, our method still produces accurate counts. Among the eight fruit types in the synthetic dataset, the **pear** (781 fruits) and **mango** (1150 fruits) scenes represent the most challenging dense-fruit configurations among the quantitatively analysed cases; our method achieves F1-scores of 0.96 and 0.95, respectively, on these scenes (Large NeRF + U-Net; see Table IV), demonstrating that the physics-guided clustering scales effectively to scenes with over a thousand densely packed fruits per tree. Qualitative results for strawberry and blueberry, under the individually-resolvable modelling conditions described in Section IV-A, are shown in Figure 6.

D. Ablation Study

In the ablation experiments, we explore the impacts of different factors on counting accuracy: reconstruction model size

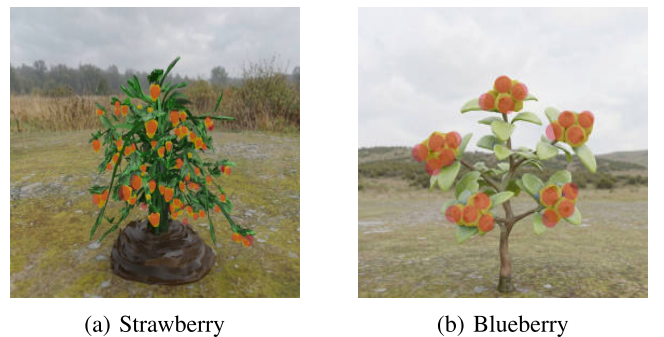


Fig. 6. Qualitative counting results for strawberry and blueberry under individually-resolvable modelling conditions. Yellow spheres = ground-truth fruit centres; red spheres = predicted cluster centres.

and segmentation strategy (Tables III and IV), the sensitivity to the regularisation parameter λ (Table V), and the effect of voxel-grid downsampling on the cost-accuracy trade-off (Table VI).

The size of the reconstruction model significantly influences counting accuracy across all datasets. We employed two reconstruction models of different sizes. The specific details

TABLE VI
EFFECT OF VOXEL GRID SIZE v ON POINT COUNT, CLUSTERING
RUNTIME, AND COUNTING ACCURACY (SYNTHETIC DATASET,
APPLE SCENE AS REPRESENTATIVE)

Voxel size v (m)	Points M'	Cluster. time (s)	F1	MAE
0.005	~185,000	98.3	0.975	13.8
0.008	~72,000	31.6	0.978	13.1
0.010 (default)	~38,000	22.4	0.977	13.3
0.015	~14,000	8.7	0.962	15.7
0.020	~7,500	4.2	0.921	19.4

are as described in the previous text, and we refer to them as Small NeRF and Large NeRF. Large NeRF consistently outperforms Small NeRF in counting metrics, particularly in F1-score and MAE. For instance, on the Mango dataset, Large NeRF with U-Net achieves an F1-score of 0.95 and MAE of 13.0, compared to Small NeRF with U-Net's F1-score of 0.93 and MAE of 14.0. Similarly, on the Lemon dataset, Large NeRF with U-Net yields an F1-score of 0.94, surpassing Small NeRF's 0.92. These results suggest that Large NeRF's enhanced capacity to capture complex scene representations improves the quality of reconstructed inputs, thereby boosting counting precision. Small NeRF, while slightly competitive in reconstruction quality on the Pear dataset, fails to match Large NeRF's counting performance, indicating that model capacity is critical for downstream tasks.

The choice of segmentation method also plays a pivotal role in counting accuracy. U-Net consistently outperforms SAM, particularly when paired with Large NeRF. On the Pear dataset, Large NeRF with U-Net achieves an F1-score of 0.96 and MAE of 12.3, significantly better than Large NeRF with SAM's F1-score of 0.90 and MAE of 16.5. This trend holds across datasets, with U-Net improving IoU scores, which directly enhance the quality of reconstructed inputs for counting. For example, on the Mango dataset, U-Net yields an IoU of 0.60 compared to SAM's 0.52, correlating with better counting metrics. U-Net's superior segmentation precision ensures more accurate object delineation, which is critical for reliable counting, whereas SAM's lower IoU limits its effectiveness.

To investigate the method's sensitivity to the regularization parameter λ , we conducted an ablation study on its initialization. Our method automatically initializes λ as the median of nearest neighbor distances, which corresponds to a scaling factor of $\beta = 1.0$. We analyzed the performance by varying this initial λ with different β factors, with the results presented in Table V.

The quantitative results demonstrate that our framework maintains high performance within a stable range around the automatic initialization. On the Synthetic and Real-World datasets, the F1-score remains robust ($F1 > 0.95$) for β values between 0.80 and 1.00. On the Synthetic dataset, the F1-score peaks at 0.977 for our proposed $\beta = 1.0$. Similarly, on the Real-World dataset, our $\beta = 1.0$ achieves the peak F1-score of 0.960. Crucially, our automatic initialization strategy ($\beta = 1.0$) consistently yields the lowest MAE (13.26 and 14.15) on these two datasets.

Performance degrades significantly only at extreme values of β . When β is set too low (e.g., $\beta \leq 0.25$), the low regularization penalty leads to severe over-segmentation. This is reflected by high Recall but poor scene-level counting accuracy (e.g., $\text{Acc} = 0.865$ on Synthetic at $\beta = 0.25$), because the resulting over-count inflates the total-count error and drives the MAE sharply upward. Conversely, when β is set too high (e.g., $\beta \geq 5.0$), the high penalty causes aggressive merging and under-segmentation. This is evidenced by a catastrophic drop in Recall (e.g., 0.640 on Synthetic and 0.400 on Fuji) and the highest MAE values across all datasets.

On the more challenging Fuji dataset, the optimal F1-score (0.931) is achieved at $\beta = 0.80$. This is slightly higher than our automatic $\beta = 1.0$ ($F1 = 0.840$), which is consistent with the dataset's high recall (0.970) at the baseline, indicating a tendency toward over-segmentation. However, $\beta = 1.0$ yields the lowest MAE (16.14) on the Fuji dataset, confirming it as the most balanced choice for count-error minimization. This analysis confirms that our automatic initialization is an effective and reliable choice, and that the energy optimization framework is stable even with moderate deviations in this parameter.

Effect of voxel-grid downsampling: The voxel-grid filter reduces the number of points entering clustering and standardises the spatial resolution, both of which affect cost and accuracy. We varied the voxel size v from 0.005 m to 0.020 m and measured the clustering runtime and the counting F1-score on the synthetic dataset (apple scene as representative; Table VI). For $v \leq 0.010$ m the F1-score stays in a narrow plateau of 0.977–0.978, and the finest resolution ($v = 0.005$) is not better (and marginally worse), because excessively dense point clouds amplify noise in PCA-based normal estimation. Beyond $v = 0.010$ m, accuracy degrades as the fine-scale geometric signatures at inter-fruit creases are averaged away within a single voxel. The default $v = 0.010$ m therefore occupies a practical sweet spot of the cost–accuracy trade-off: it is within 0.001 F1 and 0.2 MAE of the best-performing setting ($v = 0.008$ m) while reducing clustering time by more than $20\times$ relative to the raw point cloud and by $4.4\times$ relative to the finest tested resolution.

E. Computational Cost and Runtime Analysis

To assess practical deployability, we measured the wall-clock time of each pipeline stage and compared the overall cost with FruitNeRF and 2D baselines. All measurements use the hardware described in Section IV-B, with input images at 1024×1024 px (synthetic) and 1000×1500 px (real-world), averaged over three independent runs. Table VII reports the per-stage runtime, and Table VIII gives a per-image timing reference for representative 2D baselines. NeRF training dominates the full pipeline ($> 95\%$ of total time) and is shared between FruitNeRF and our method. Our physics-guided clustering adds a 4-second overhead in the clustering sub-step (22 s vs. 18 s) but saves 5 s in the counting sub-step ($2.7\times$ speed-up: 3 s vs. 8 s, due to the parameter-free volume-based counting replacing template matching), for a net 1-second reduction in the post-NeRF stage (72 ± 3 s vs. 73 ± 3 s for FruitNeRF). Relative to the total pipeline budget, this

TABLE VII

PER-STAGE RUNTIME (MEAN $\pm$ STD). NeRF TRAINING TIMES ARE IDENTICAL BECAUSE THE BACKBONE IS UNCHANGED. POST-NeRF TOTAL IS THE PRACTICALLY RELEVANT RUNTIME FOR REPEATED YIELD ESTIMATION

Stage	FruitNeRF [3]	Ours
<i>NeRF Training (one-time offline cost)</i>		
NeRF Training (Small)	12.1 $\pm$ 0.3 min	12.1 $\pm$ 0.3 min
NeRF Training (Large)	148.7 $\pm$ 2.1 min	148.7 $\pm$ 2.1 min
<i>Post-NeRF Processing (repeated stage)</i>		
Point Cloud Export	45 $\pm$ 2 s	45 $\pm$ 2 s
Preprocessing (downsample + denoise)	2 $\pm$ 0.1 s	2 $\pm$ 0.1 s
Clustering	18 $\pm$ 1 s	22 $\pm$ 1 s
Counting	8 $\pm$ 0.5 s	3 $\pm$ 0.2 s
Post-NeRF Total	73 $\pm$ 3 s	72 $\pm$ 3 s

TABLE VIII

INFERENCE-TIME REFERENCE FOR REPRESENTATIVE 2D BASELINES (PER IMAGE). THESE METHODS ARE FAST PER IMAGE BUT REQUIRE ADDITIONAL TRACKING OR DEDUPLICATION TO HANDLE MULTI-VIEW COUNTING

Method	Per-image Inference	Multi-view Deduplication
GMN [34]	~ 0.2 s	Manual / not supported
FamNet [35]	~ 0.4 s	Manual / not supported
CounTR [36]	~ 0.3 s	Manual / not supported

TABLE IX

F1-SCORE UNDER THREE CONTROLLED DENSITY CONDITIONS (AVERAGED ACROSS EIGHT FRUIT TYPES, MEAN $\pm$ STD)

Method	Sparse (> 5 cm)	Medium (2–5 cm)	Dense (< 2 cm)
FruitNeRF [3]	0.852 $\pm$ 0.031	0.755 $\pm$ 0.048	0.643 $\pm$ 0.071
Ours	0.988 $\pm$ 0.012	0.977 $\pm$ 0.019	0.966 $\pm$ 0.033

change is well under 0.1%, meaning the substantial accuracy improvement comes at virtually no additional computational cost. Compared with 2D baselines, per-image inference is faster (~ 0.2 – 0.4 s) but those methods do not inherently resolve multi-view duplicate counting, which our 3D framework eliminates by construction.

V. DISCUSSION

In this section we interpret the results beyond the raw numbers: how performance varies with fruit density, how the framework behaves under different lighting conditions, and what the two representative failure modes reveal about the operational envelope of the method.

A. Performance Across Fruit Density Levels

Table IX reports F1-scores under the three controlled density conditions in the synthetic dataset, averaged across all eight fruit types. These three per-density averages are weighted across the synthetic dataset's fruit-type distribution to recover the Synthetic F1 of 0.75 (FruitNeRF) and 0.977 (Ours)

TABLE X

COUNTING PERFORMANCE UNDER FOUR CONTROLLED LIGHTING CONDITIONS (SYNTHETIC DATASET, APPLE SCENE AS REPRESENTATIVE). HERE, ACC DENOTES SCENE-LEVEL COUNTING ACCURACY $1 - |N_{\text{pred}} - N_{\text{gt}}|/N_{\text{gt}}$

Lighting Condition	Acc	Recall	F1	MAE
Normal daylight	0.996	0.954	0.977	13.26
Strong sunlight	0.988	0.941	0.964	14.53
Diffuse / overcast	0.985	0.937	0.960	14.87
Backlight	0.971	0.921	0.945	16.74

reported in Table II. The performance gap over FruitNeRF widens as density increases: at sparse density both methods perform well, but at dense density (physical contact, inter-fruit distance < 2 cm) FruitNeRF's F1 drops to 0.643 while ours remains at 0.966. The reason is structural rather than parametric. When two fruits are in contact, their Euclidean separation is zero, so no distance threshold—regardless of tuning—can separate them, which is why DBSCAN-based pipelines collapse in this regime. Our physics-guided approach exploits the fact that the normal field still exhibits a sharp discontinuity at the contact crease: $\arccos(\kappa(\mathbf{n}_i \cdot \mathbf{n}_j))$ approaches π at the junction of two convex surfaces, which places a large factor in the denominator of Eq. (7) and suppresses θ precisely at the boundary, enforcing a segmentation cut. The high F1 at dense density is therefore a genuine algorithmic advantage, not the product of parameter tuning.

B. Robustness to Lighting Conditions

In real-world orchards, the relative geometry between camera, sun, and fruit surface changes continuously, producing direct illumination, partial shadow, and specular highlights on the same fruit across different frames. Our framework absorbs these effects through the structural design of the NeRF components: the view-conditioned Appearance Field absorbs specular highlights and directional shadows, while the position-only Semantic Field produces consistent fruit logits regardless of lighting angle. The physics-guided clustering operates exclusively on geometry, further insulating the final count from any residual photometric effects. To quantify this robustness, we configured four controlled lighting scenarios in the Blender synthetic environment (Figure 7); counting performance under each is reported in Table X. Accuracy remains consistently high across all four conditions, with the moderate decrease under backlight ($\Delta F1 = 0.032$) attributable to the reduced quality of upstream segmentation masks in high-contrast silhouette regions. Under truly extreme conditions (near-darkness, severe over-exposure), photometric supervision becomes uninformative and reconstruction quality will degrade, but such conditions fall outside the normal operating envelope of field data acquisition.

C. Failure Cases and Applicability Scope

Two representative failure modes delineate the operational envelope of the method.

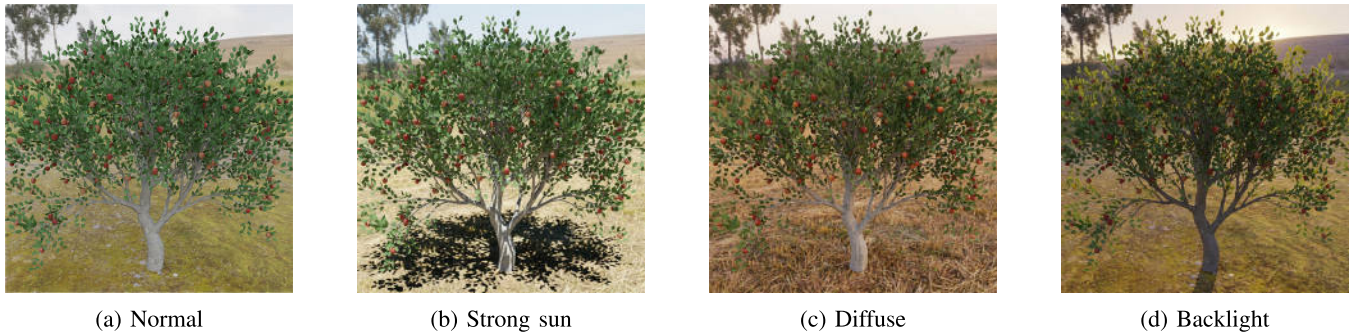


Fig. 7. Representative input images under four common orchard lighting conditions: (a) normal daylight, (b) strong direct sunlight, (c) weak/diffuse light, (d) backlight.



Fig. 8. Failure case 1: Fuji-SfM Tree 3. A constrained linear camera trajectory produces incomplete point clouds at the canopy periphery (red = predicted cluster centres, yellow = ground-truth centres).

Case 1: insufficient hemispherical coverage. Tree 3 of the Fuji-SfM dataset lies at the end of a multi-tree row, where the capture trajectory is constrained to a partial arc rather than a complete hemispherical orbit. Fruits at the canopy periphery are observed from a limited angular range, yielding incomplete point clouds and unreliable normal estimates. The clustering stage consequently under-segments fruit groups at the canopy extremities (Figure 8). This is fundamentally an acquisition-geometry problem: when the same tree is re-scanned with a complete orbit, accuracy recovers to the level of Trees 01–02. This motivates acquisition-protocol guidance (minimum orbit arc and minimum number of viewpoints per fruit) as future work.

Case 2: permanently occluded cluster-internal fruits. In dense blueberry clusters (Figure 9), inner berries are completely surrounded by outer layers and remain invisible from any external viewpoint. The NeRF converges to a single opaque outer shell, and the point cloud lacks the per-berry geometry that our clustering needs. This is a fundamental limitation shared by *all* multi-view reconstruction-based counting pipelines: invisible fruits cannot be reconstructed from RGB observations alone. Overcoming this requires complementary sensing modalities or generative priors about bunch structure.

Based on these cases, the framework is applicable to fruits satisfying two conditions: (C1) approximate convexity, so that normals reverse at inter-fruit contact boundaries; and (C2)



Fig. 9. Failure case 2: densely occluded blueberry cluster. Inner berries that are never visible from any viewpoint cannot be reconstructed; the NeRF yields only the outer shell, which is correctly recognised as a single convex surface and thus counted as one instance (red = predicted cluster, yellow = ground-truth berry centres).

individual resolvability, i.e., each fruit is at least partially visible from the camera trajectory. Fruits satisfying both include apples, pears, lemons, peaches, mangoes, plums, and tomatoes (including those in physical contact on a truss). Fruits violating (C2) due to cluster-internal occlusion—such as inner berries in grape bunches or tightly packed blueberry clusters—are outside the applicability scope of our current framework. Additionally, the NeRF assumption of scene rigidity is violated by wind-induced motion of branches and fruit; robust-loss formulations such as RobustNeRF [45] are a straightforward extension to mitigate this in future work.

VI. CONCLUSION

This paper introduces a dual-framework that integrates 3D neural reconstruction and physics-guided adaptive clustering to address object counting in occluded, clustered 3D scenes. To address the limitations of 2D methods that suffer from duplicate counting when aggregating multi-view observations, we employ a semantic NeRF to reconstruct the scene and segment target instances simultaneously. This approach eliminates background interference and view-based duplicates without requiring extensive 3D annotations. To overcome the challenges of traditional clustering algorithms, which are often physics-agnostic and rely on rigid

thresholds, we propose a physics-guided clustering method that adapts to object morphology. This algorithm fuses multi-dimensional geometric features, including surface normals and elevation gradients, and utilizes a hierarchical optimization framework to avoid manual parameter tuning. We integrate the semantic reconstruction and adaptive clustering into a unified pipeline for robust counting. Extensive experiments on the synthetic, real-world, and Fuji-SfM datasets demonstrate that our approach improves counting accuracy by 6.9 percentage points (0.996 vs. 0.927) over the state-of-the-art 3D baseline on the synthetic dataset, with corresponding gains of 7.2 pp and 12.9 pp on the real-world and Fuji datasets. For future work, we aim to generalise the framework to more diverse crops, adopt faster reconstruction backbones such as 3D Gaussian Splatting for real-time deployment, and integrate multi-modal sensors to lift the individual-resolvability requirement in cluster-structured fruits such as grapes and dense blueberries.

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Yimo Wang received the B.Eng. degree from Southeast University, Nanjing, China, in 2023, where he is currently pursuing the Ph.D. degree in control science and engineering with the School of Automation. His research interests include deep learning and pattern recognition.



Bin Kang (Member, IEEE) received the Ph.D. degree in electrical engineering from Nanjing University of Posts and Telecommunications, Nanjing, China, in 2016. He is an Associate Professor with Nanjing University of Posts and Telecommunications. His research interests are in the area of computer vision and pattern recognition.



Jian Liu (Member, IEEE) received the B.S. and Ph.D. degrees from the School of Automation and Electrical Engineering, University of Science and Technology Beijing, Beijing, China, in 2015 and 2020, respectively. From September 2017 to 2018, he was a Joint Training Student with the Department of Mathematics, Dartmouth College, Hanover, NH, USA. From 2020 to 2021, he was a Post-Doctoral Fellow with the School of Automation, Southeast University, Nanjing, China, where he is currently an Associate Professor. His research interests include multiagent systems, nonlinear control, event-triggered control, fixed-time control, and intermittent control.



Changyin Sun (Senior Member, IEEE) received the B.S. degree in mathematics from Sichuan University, Chengdu, China, and the M.S. and Ph.D. degrees in electrical engineering from Southeast University, Nanjing, China, in 2001 and 2004, respectively. He is currently a Professor with the School of Artificial Intelligence, Anhui University, Hefei, China. His research interests include artificial intelligence, intelligent control, and optimal theory. He is an Associate Editor of *IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS* and *IEEE/CAA JOURNAL OF AUTOMATICA SINICA*.